Supplementary Material: Diagnosing Prototype Estimation in Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification

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Roadmap. We first specify information access, notation, datasets, uncertainty, and the counterfactual diagnostics. We then report support-only selection, ablations, complete matrices, implementation provenance, resource measurements, query-assisted robustness, and the base-only representation control. Machine-readable manifests and verification records accompany the code artifact.

A. Protocol and Reproducibility Details

All quantitative result tables are generated from validated result bundles rather than transcribed by hand. Before generation, the validator checks row uniqueness, numeric ranges, the presence of all twenty support draws, matched feature and protocol identifiers, support indices, and selection–safety–target provenance. Projected RanPAC must also satisfy a scale-free normal-equation residual test. The cross-family diagnosis, support-only, QPR, cumulative-calibration, and query-self-exclusion bundles pass the same checks.

Throughout, the *safety partition* is the held-out base audit split used only after selection. Generated tables may abbreviate the corresponding measurement as "Audit gain."

Query-local implementations are tested for invariance to query order and computational batching. Two of 10,000 predictions change under chunking: released ULA on CIFAR-100 and released TEEN on miniImageNet. Reproduction gives maximum score changes of 3.58×10^{-5} and 7.75×10^{-7}, respectively, with winning margins below 1.6×10^{-5}. These are floating-point near ties; neither method aggregates queries. The transductive benchmark uses the equation-level ProtoLP reconstruction described in Sec. G.

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A.1. Method access and optimization

Table 1 records the information available to each method. Here, "unlabeled cohort" means that a prediction may change when other test images are added or removed. This is distinct from computationally batching independent support-only scores. Methods that use a query-class prior, text, supervised optimization, or full novel labels are not mixed into the primary support-only rank.

The HyCal entry denotes only our algebraic reduction of its published standard five-shot setting $(\alpha, \beta) = (1, 0)$, with $\beta = 0$ interpreted as the intended hard-threshold limit. It is not a claim that general HyCal is image-only.

A.2. Notation

Table 2 distinguishes population quantities, estimates obtained from base data, and quantities available only for retrospective analysis or query-assisted inference.

A.3. Representation provenance and novelty scope

Table 3 states what the released checkpoint documentation can and cannot establish. OpenAI reports CLIP training on 400M web image–text pairs [10], but does not release the corpus [8]; exact benchmark membership is therefore unknown. The MAE and supervised ResNet checkpoints use ImageNet-1K, which directly overlaps the source of miniImageNet. CUB-200-2011 documentation warns of ImageNet image overlap [11]; exact checkpoint membership remains unaudited. Accordingly, all released-encoder experiments establish only that a class is new to the incremental classifier. We make no pretraining-novel claim.

A.4. Datasets, supports, and selection protocol

The CUB protocol is $100 + 10 \times 10$ classes; CIFAR-100 and miniImageNet are $60 + 8 \times 5$; Aircraft is $50 + 10 \times 5$.

Table 1. Information and optimization used by the evaluated method families. A dash means that the source is unavailable. Support-only and query-assisted methods are ranked separately.

Method family	Fitting or base information	Labeled supports	Unlabeled cohort	Additional input	Evaluation role
NCM, LDA, TEEN, ULA, FeCAM, HyCal reduction, RanPAC, RanDumb, ShrinkSel	Closed-form base statistics and/or base-only parameter selection	Yes	No	—	Primary support-only comparison
CLIP-Adapter [2] and ADAM/APER reductions [15]	Supervised base-image optimization once	Yes	No	—	Matched adaptation comparison
Joint linear probe	Supervised head fit at every session with base replay	Yes	No	—	Matched optimization comparison
TIM, LaplacianShot, ProtoLP, ShrinkSel+QPR	Frozen head and base-only parameter selection	Yes	Yes	—	Marginal-free query-assisted comparison
MAP-OT$_u$ (PT-MAP core)	Frozen head and base-only parameter selection	Yes	Yes	Uniform prior	Prior-assisted comparator
TIM++ [5]	CLIP zero-shot/text initialization	Yes	Yes	Text	Multimodal context
Known-marginal decoder	Frozen selected head	Yes	Yes	Exact proportions	Oracle diagnostic only
Full prototype / linear probe	Complete labeled training split	Full	No	Full novel labels	Retrospective counterfactual only
FDR [14]	Frozen CLIP; optional base-session LoRA	Method-specific	No	Text	Multimodal context

Table 2. Notation used throughout the method. T is the final session index (and the number of incremental sessions), whereas T_c is a labeled image set for class c.

Symbol	Meaning	Information boundary or role
$t \in \{0, \dots, T\}$; D_t, Y_t	Session index; labeled data and disjoint classes introduced at that session	$t = 0$ is the base session; $t > 0$ supplies K supports per new class
$\mathcal{C}_t = \bigcup_{j=0}^{t} Y_j$	All classes seen through session t	Candidate label set at evaluation
$f_{\text{img}}, z(x), D$	Frozen visual encoder, normalized image feature, and feature dimension	No downstream class name, prompt, or text embedding enters z
S_c^K, T_c	Available K-shot support features and complete labeled training features for class c	Complete base sets are available for fitting; complete novel sets are restricted to retrospective analysis
$\hat{\mu}_c^K, \hat{\mu}_c^{\text{full}}$	Normalized K-shot and complete-data prototype estimates	Matched cosine decision rule with different class-mean information
$A_n^K, A_n^{\text{full}}, A_n^{\text{LP}}, r$	Novel accuracy for the K-shot prototype, full prototype, and full-data linear probe; ordered full-mean share	Accounting quantities, not deployable-method inputs
μ_c, Σ; $\bar{\Sigma}_B, \delta, M$	Population class mean/shared covariance; pooled base covariance, mean eigenvalue, and within-class degrees of freedom	First pair is analytical; second triplet is estimated from complete base-session features only
$\mathcal{H}, h, p_c^h$	Candidate-head family; one candidate $h = (p, \lambda)$; its prototype vector for class c	Unit, raw, or finite-shot-corrected radius crossed with the covariance-shrinkage grid
$\Lambda, \lambda, \gamma, G_\lambda$	Shrinkage grid, selected shrinkage, fixed ridge multiplier, and comparison metric	Selected from class-disjoint base episodes; frozen on target sessions
B, m, w_b, w_n, J	Base-class count, per-session class increment, schedule-derived base/novel weights, and selector objective	Derived without target arrays
$\mathcal{U}_t, \mathcal{C}_n$	Unlabeled query cohort and incremental classes seen by session t	Available only to query-assisted methods; no target labels or explicit class-proportion vector enter inference
$\kappa, \tau_q, m_{\min}, \alpha$	Logit scale, novel-mass gate, minimum admitted count, and rectification step	QPR tuple selected and safety-checked on base episodes

Every incremental class has five labeled supports. Support seeds are exactly $1, \dots, 20$. Within each class, the standard five-shot draw is a prefix of the fixed sequence used for $K \in \{1, 2, 5, 10, 20\}$. The three standard datasets use identical support indices across the CLIP, MAE-ft, and supervised ResNet-50 encoders; complete protocol records accompany the code.

The DTD [1] protocol is $27 + 4 \times 5$ classes on official

Table 3. Documented representation training source and the novelty claim supported by each encoder–benchmark group. Unknown membership is not interpreted as absence.

Representation and benchmark	Documented training source	Exposure status	Supported interpretation
CLIP × all five benchmarks	WIT-400M web image–text data; corpus unreleased	Exact images and semantic coverage cannot be audited	Classifier-new classes only
MAE-ft ViT-B/16 × mini-ImageNet	MAE pretraining and supervised fine-tuning on ImageNet-1K	Direct source-dataset and class overlap	Classifier-new classes only
MAE-ft ViT-B/16 × CUB/CIFAR	MAE pretraining and supervised fine-tuning on ImageNet-1K	Exact checkpoint-training membership unaudited	Classifier-new classes only
Supervised ResNet-50 × miniImageNet	Supervised ImageNet-1K training	Direct source-dataset and class overlap	Classifier-new classes only
Supervised ResNet-50 × CUB/CIFAR	Supervised ImageNet-1K training	Exact checkpoint-training membership unaudited	Classifier-new classes only
Base-only ResNet-18 × standard benchmarks	Training from scratch on base-session images only	Novel-class training images and labels are excluded	Benchmark-class-disjoint control

Table 4. Architecture and feature dimension of the additional frozen encoders.

Encoder	Feature dimension D
MAE-ft ViT-B/16	768
Sup. IN1K ResNet-50	2048

split 1. A seed-1993 permutation defines the order of the 47 semantic classes; train1+val1 supplies 80 labeled images per class and test1 supplies 40 queries. The official r1.0.1 archive, annotations, complete class permutation, and split manifests are recorded in the released artifacts. Its twenty deterministic support draws use the same nested $K \in \{1, 2, 5, 10, 20\}$ construction as the other standalone extension.

Base classes are permuted once with protocol seed 20260812 and split 60%/20%/20% by class into construction, selection, and safety sets. This yields 60/20/20 classes for CUB, 36/12/12 for CIFAR-100 and miniImageNet, and 30/10/10 for Aircraft. The corresponding DTD split is 16/5/6 classes. For the support-only ShrinkSel selector, construction and selection identities are pooled, deterministically shuffled with seed 8675309, and repartitioned into four folds. In each fold, one class subset acts as pseudo-novel and the remaining pooled identities act as anchors; the safety identities remain entirely excluded. The selector allocates 200 episodes across these folds, and the selected candidate is then checked on 200 independent safety episodes.

QPR retains the original roles instead: construction identities provide anchors and selection identities provide disjoint pseudo-novel classes. It uses 20 balanced and 20 deterministic-imbalanced episodes for selection and the same counts on safety. Each sampled pseudo-novel class contributes five supports and at most fifteen queries. Both groups use the prototype treatment selected by the parent ShrinkSel head. Projected RanPAC uses twenty base episodes because its 10k-dimensional solve is substantially more expensive. Target arrays are absent from all selector data structures.

B. Additional Diagnostic Analysis

Tables 5 and 6 report the complete matched accounting behind the main counterfactual-path figure. All three accuracies in a row use the same frozen representation and evaluation classes; the two full-data columns also use the same complete labeled training images.

B.1. Linear-probe and bootstrap details

The counterfactual multinomial probe uses scikit-learn logistic regression with inverse-regularization parameter $C = 1$, ℓ_2 penalty, an intercept, no class reweighting, LBFGS, tolerance 10^{-6}, and at most 2,000 iterations. It is fit to every seen-class training feature and never changes the feature extractor. All sixteen runs converge without a warning (33–139 LBFGS iterations, 59 for Aircraft).

For one two-way bootstrap draw, we sample twenty support-draw indices with replacement and C_n novel-class indices with replacement. A_n^K is averaged over the resulting crossed $20 \times C_n$ matrix. A_n^{full} and A_n^{LP} are averaged over the same resampled classes. We then recompute the two accounting terms and r. Five thousand draws form percentile intervals. This propagates both sources of variation; a seed-only interval would treat the evaluated novel classes as fixed. We report r only when the 95% interval for its denominator, $A_n^{\text{LP}} - A_n^K$, excludes zero. Because the path first replaces the finite- shot means and then replaces the classifier, the two terms and r are ordered and path-dependent; neither is a unique causal attribution.

Table 5. Matched-feature novel-class accounting on frozen CLIP image features (20 support seeds). A_n^K, A_n^{full}, and A_n^{LP} use the same cached representation. Mean Δ is $A_n^{\text{full}} - A_n^K$; LP resid. is $A_n^{\text{LP}} - A_n^{\text{full}}$; r is their ordered full-mean share, with a two-way-bootstrap 95% interval over support draws and novel classes. All values except r are percentage points.

Backbone	Dataset	A_n^K	A_n^{full}	A_n^{LP}	Mean Δ	LP resid.	r [95% CI]
ViT-B/16	CUB-200	57.34	75.79	73.45	+18.45	-2.34	1.15 [1.02, 1.31]
ViT-B/16	CIFAR-100	45.53	68.82	77.97	+23.29	+9.15	0.72 [0.67, 0.76]
ViT-B/16	miniImageNet	79.76	88.97	94.05	+9.22	+5.07	0.64 [0.57, 0.72]
ViT-B/16	FGVC-Aircraft	25.74	49.21	50.54	+23.46	+1.34	0.95 [0.88, 1.03]
ViT-B/32	CUB-200	45.04	66.48	64.55	+21.45	-1.93	1.10 [1.00, 1.22]
ViT-B/32	CIFAR-100	40.34	64.57	75.07	+24.23	+10.50	0.70 [0.65, 0.74]
ViT-B/32	miniImageNet	76.14	86.40	92.35	+10.26	+5.95	0.63 [0.56, 0.71]
RN50	CUB-200	39.29	60.74	59.46	+21.45	-1.28	1.06 [0.97, 1.18]
RN50	CIFAR-100	24.49	45.47	61.17	+20.98	+15.70	0.57 [0.53, 0.62]
RN50	miniImageNet	70.81	83.30	90.87	+12.49	+7.57	0.62 [0.56, 0.68]

Table 6. Cross-family matched-feature accounting over 20 support draws. MAE-ft denotes MAE pretraining followed by the released ImageNet-1K fine-tuning checkpoint. Classes are new to the downstream classifier; exclusion from representation pretraining is not claimed.

Backbone	Dataset	A_n^K	A_n^{full}	A_n^{LP}	Mean Δ	LP resid.	r [95% CI]
MAE-ft ViT-B/16	CUB-200	43.07	60.48	66.53	+17.41	+6.05	0.74 [0.69, 0.79]
MAE-ft ViT-B/16	CIFAR-100	51.18	73.77	82.40	+22.59	+8.62	0.72 [0.68, 0.77]
MAE-ft ViT-B/16	miniImageNet	98.19	98.95	98.85	+0.76	-0.10	1.15 [0.81, 3.02]
IN1K RN50	CUB-200	33.86	51.14	55.02	+17.28	+3.88	0.82 [0.75, 0.89]
IN1K RN50	CIFAR-100	31.45	57.02	71.02	+25.58	+14.00	0.65 [0.60, 0.69]
IN1K RN50	miniImageNet	90.63	95.85	97.50	+5.22	+1.65	0.76 [0.60, 0.85]

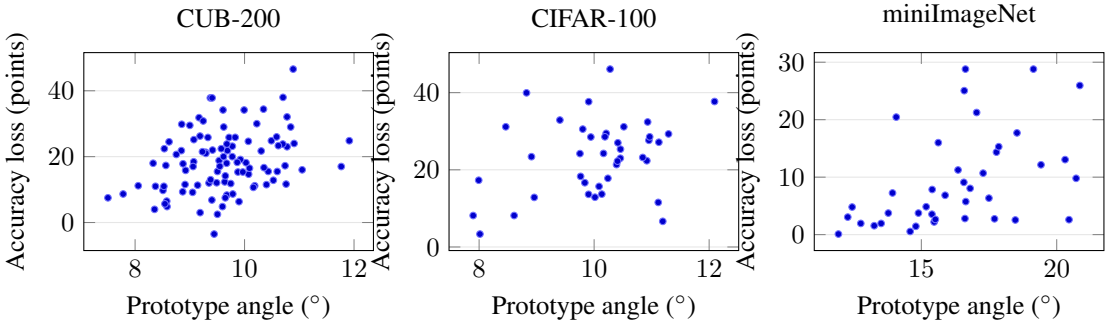


Figure 1. Five-shot prototype angle versus classwise accuracy recovered by replacing the support mean with the complete-data mean for CLIP ViT-B/16. Points are novel class–support pairs. Direction error is informative on the three standard datasets, but its effect on a decision also depends on the competing-class margins.

Table 7 makes the full-mean replacement term concrete. "Repaired" means wrong under the five-shot prototype and correct under the full prototype; "harmed" is the reverse. Counts are means over twenty supports and include only final-session novel test examples. Repairs exceed harms for every pair. Aircraft is the exception to the classwise angle relationship, not to the aggregate repair result.

B.2. Analytical penalty versus empirical error

For raw sample means under independent sampling, the shared-covariance approximation predicts $\mathbb{E}\|\hat{\mu} - \mu\|_G^2 = \text{tr}(G\Sigma)/K$. Table 8 compares this value with the measured squared metric error to the complete-training mean at $K = 5$. The table retains the covariance value selected by the normalized-only ablation so that raw and unit prototypes can be compared under an identical metric; it does not list the final joint selector. The raw empirical value closely follows the prediction on CIFAR-100, is lower on CUB (partly due to

Table 7. Prototype-angle correlation with classwise accuracy loss and mean decision transitions after substituting full novel means.

Dataset	Pearson	Spearman	Repaired	Harmed
ViT-B/16				
CUB-200	0.51	0.50	637.8	101.3
CIFAR-100	0.54	0.52	1019.1	87.3
miniImageNet	0.49	0.47	402.9	34.3
FGVC-Aircraft	-0.10	-0.07	429.2	39.6
ViT-B/32				
CUB-200	0.47	0.47	748.2	121.2
CIFAR-100	0.44	0.39	1055.2	85.8
miniImageNet	0.50	0.43	455.2	44.7
RN50				
CUB-200	0.47	0.47	753.1	127.0
CIFAR-100	0.41	0.38	972.5	133.3
miniImageNet	0.49	0.49	559.8	60.0

finite-population sampling), and is higher on miniImageNet. Normalizing a random sample mean is nonlinear and generally adds discrepancy, especially for miniImageNet/RN50. Thus the analysis explains scaling and the metric trade-off; it is not used as a calibrated equality for normalized features.

Table 8. Five-shot analytical penalty and empirical prototype error under the normalized-only ablation's selected metric. Theory is $\mathrm{tr}(G\Sigma)/5$; Raw and Normalized are empirical squared metric errors. These λ values are not the final joint selections.

Dataset	$\lambda^\star$	Theory	Raw	Normalized
ViT-B/16				
CUB-200	0.75	34.9	29.1	33.4
CIFAR-100	0.75	36.7	36.6	41.7
miniImageNet	0.50	37.9	46.0	63.6
ViT-B/32				
CUB-200	0.75	34.7	28.7	33.7
CIFAR-100	0.50	34.2	33.7	39.3
miniImageNet	0.50	38.0	45.2	63.5
RN50				
CUB-200	0.75	52.9	44.5	53.2
CIFAR-100	0.90	65.5	65.2	74.9
miniImageNet	0.75	64.1	78.1	117.3

The theoretical column is exactly proportional to $1/K$. Empirically, raw and normalized metric errors also fall strongly from $K = 1$ to 20, although comparison to the finite complete-training mean introduces a finite-population correction at large K. Figure 2 gives the corresponding descriptive downstream view: cosine-NCM novel accuracy rises monotonically on all nine dataset–encoder pairs, with the largest changes between one and five shots. This accuracy trend is consistent with, but does not by itself establish, the analytical $1/K$ error law. No target error value is fed back into selection.

C. Support-Only Ablations and Selection Stability

The final selector crosses three prototype treatments—unit, raw, and finite-shot-corrected—with seven covariance-shrinkage values. At $K = 1$, corrected is defined to equal raw; the squared-resultant correction is applied only when at least two samples are available. Table 9 reports the selected candidate, its worst class-disjoint selection-fold gain over raw shared LDA, and its held-out safety-set gain. Nonnegative worst-fold gain is a hard feasibility constraint. The safety criterion permits at most a 0.10-point regression; failure would deploy raw $\lambda = 0$ shared LDA instead. ViT-B/32/CIFAR-100 records -0.04 and therefore passes, while every other core row improves. Aircraft selects the raw $\lambda = 0$ endpoint and exactly equals LDA.

Table 9. Robust base-only prototype-and-geometry selection. "Corrected" is the finite-shot resultant-length correction defined in the main paper. Both reported gains are relative to raw-mean shared LDA and use base data only.

Dataset	Prototype	$\lambda^\star$	Worst-fold gain	Audit gain
ViT-B/16				
CUB-200	corrected	0.75	+0.68	+0.64
CIFAR-100	corrected	0	+0.08	+0.13
miniImageNet	corrected	0	+0.21	+0.28
ViT-B/32				
CUB-200	corrected	0.5	+0.90	+1.01
CIFAR-100	corrected	0.1	+0.18	-0.04
miniImageNet	corrected	0.25	+0.14	+0.21
RN50				
CUB-200	unit	0.75	+1.32	+1.83
CIFAR-100	corrected	0	+0.07	+0.08
miniImageNet	corrected	0.5	+0.70	+0.49
ViT-B/16				
FGVC-Aircraft	raw	0	+0.00	+0.00

Table 10 reports the complete sixteen-setting comparison with shared LDA. All fifteen non-Aircraft intervals are positive, while Aircraft selects the exact LDA endpoint. Exact two-sided sign-flip tests with Holm correction retain all nine core contrasts; the three core CIFAR effects nevertheless remain below the stated .10-point practical threshold. Table 11 adds the nonredundant Base/Novel decomposition: every core Full gain exchanges a small base decrease for a novel increase, while the equal-weight harmonic mean rises

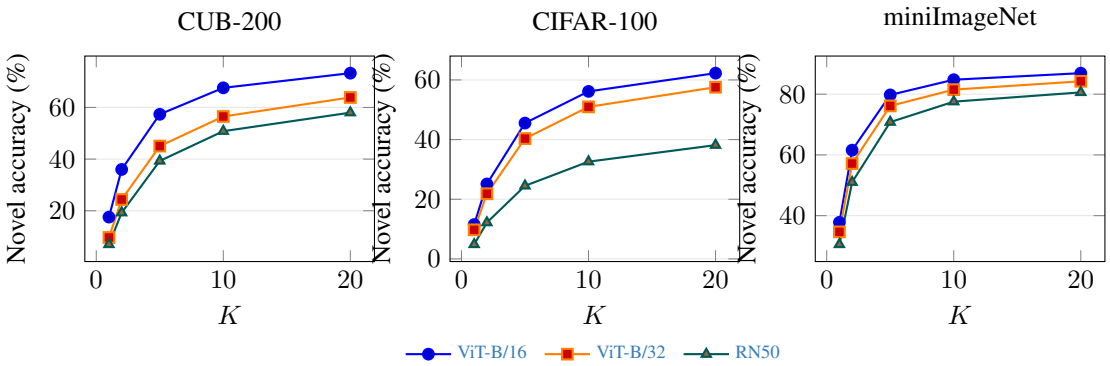


Figure 2. Cosine-NCM novel accuracy as nested support sets grow from one to twenty examples. Each point averages twenty matched draws; base prototypes and frozen features are unchanged. The curves provide a downstream shot-sensitivity check rather than a direct fit of the analytical $1/K$ metric-error law.

Table 10. Scope of the support-only result. Each entry is the paired session-average change, ShrinkSel minus shared LDA, with a 95% bootstrap interval over twenty matched support draws. Aircraft selects the exact LDA endpoint. †Effects below 0.10 point are statistically nonzero but practically tiny by the stated convention.

Dataset	Δ vs. shared LDA [95% CI]
CLIP ViT-B/16	
CUB-200	+0.27 [+0.18, +0.35]
CIFAR-100	+0.06 [+0.05, +0.08]†
miniImageNet	+0.29 [+0.27, +0.32]
Aircraft	LDA endpoint
CLIP ViT-B/32	
CUB-200	+0.49 [+0.43, +0.56]
CIFAR-100	+0.05 [+0.03, +0.08]†
miniImageNet	+0.30 [+0.26, +0.33]
CLIP RN50	
CUB-200	+1.17 [+1.05, +1.28]
CIFAR-100	+0.04 [+0.02, +0.06]†
miniImageNet	+0.66 [+0.61, +0.71]
MAE-ft ViT-B/16	
CUB-200	+0.90 [+0.86, +0.93]
CIFAR-100	+0.78 [+0.70, +0.86]
miniImageNet	+0.65 [+0.61, +0.68]
IN1K RN50	
CUB-200	+0.74 [+0.60, +0.88]
CIFAR-100	+0.50 [+0.46, +0.53]
miniImageNet	+1.52 [+1.43, +1.61]

Interpretation. All fifteen non-Aircraft paired session-average intervals are positive, but the three CIFAR effects in the core CLIP matrix are below the 0.10-point practical threshold. Aircraft selects the exact LDA endpoint. The safety construction therefore prevents a large regression without promising a meaningful gain on every target.

in every case.

C.1. Novel-order and class-balanced optimization controls

The published base/novel identity split, features, base-only head, query manifests, and twenty support draws are fixed while five deterministic permutations change only the arrival order of the already designated novel identities. The resulting crossed interval therefore measures sensitivity to novel order conditional on the published split; it does not estimate uncertainty over alternative base/novel partitions.

Every one of the five permuted orders retains a positive ShrinkSel–shared-LDA gain. The crossed intervals also remain above zero, although CIFAR-100 stays below the predeclared .10-point practical-size threshold. This supports robustness to arrival order conditional on the fixed BiMC base/novel split; it says nothing about alternative identity partitions.

The class-balanced probe uses the same complete base replay, accumulated labeled novel supports, zero initialization, AdamW optimizer, learning rate .01, weight decay 10^{-4}, and 300 full-batch steps as the ordinary joint control. Its only change is inverse-frequency class weights, which give every seen class equal total loss weight; neither version uses target queries for fitting or selection. The adapter variants optimize once on the base-session training images, whereas the joint linear probe revisits those images at every session. Their fixed choices are evaluated on held-out base classes before target evaluation.

Class balancing raises the probe's novel accuracy relative to the ordinary sample-weighted probe, but it does not uniformly dominate the closed-form head. ShrinkSel remains higher in session average on CUB-200 and miniImageNet and lower on CIFAR-100; its final novel accuracy is higher on all three. The CIFAR result shows why the optimization control cannot be summarized by novel accuracy alone: a larger base score reverses the sample-weighted session-average comparison.

Table 11. Final-session support-only group accuracy on the fixed BiMC class order. Full is sample-weighted over all seen queries; Base and Novel are group accuracies; H is the equal-weight harmonic mean of Base and Novel, computed per support draw and then averaged. Values are means over 20 matched support draws. The table exposes whether a Full-accuracy gain reallocates errors between the two groups.

		Shared LDA				ShrinkSel			
Backbone	Dataset	Full	Base	Novel	H	Full	Base	Novel	H
ViT-B/16	CUB-200	72.20	84.54	60.15	70.28	72.72	83.41	62.28	71.31
ViT-B/16	CIFAR-100	68.37	81.25	49.06	61.17	68.50	81.22	49.41	61.43
ViT-B/16	miniImageNet	88.49	95.03	78.67	86.08	88.97	94.98	79.96	86.82
ViT-B/32	CUB-200	62.93	79.08	47.14	59.06	63.93	78.35	49.84	60.92
ViT-B/32	CIFAR-100	63.97	78.65	41.95	54.70	64.13	78.43	42.68	55.26
ViT-B/32	miniImageNet	84.90	91.97	74.29	82.19	85.55	91.60	76.46	83.35
RN50	CUB-200	56.54	77.83	35.73	48.96	59.37	75.79	43.31	55.11
RN50	CIFAR-100	49.90	68.09	22.61	33.94	49.96	68.06	22.81	34.16
RN50	miniImageNet	80.77	90.66	65.94	76.34	82.15	89.86	70.59	79.06

Table 12. Novel-arrival-order robustness with the base/novel split fixed. Published is the original session-average ShrinkSel–shared-LDA gain. Permuted results average five deterministic novel orders and 20 supports; brackets are a two-way bootstrap over both axes. This does not test alternative base/novel splits.

Dataset	Orig.	Permuted [CI]	Range	> 0
CUB-200	+0.27	+0.21 [+0.11, +0.31]	+0.15–+0.27	5/5
CIFAR-100	+0.06	+0.07 [+0.06, +0.09]	+0.07–+0.08	5/5
miniImageNet	+0.29	+0.27 [+0.24, +0.30]	+0.24–+0.32	5/5

C.2. Radius–geometry factorial ablation

Table 15 isolates the two selected factors. The "geometry only" arm uses unit prototypes with the selected λ; "radius only" uses the selected prototype treatment at the cosine endpoint; and "full covariance" uses the selected treatment at $\lambda = 0$. This design matters because comparing only against cosine would conflate covariance geometry with prototype length.

The radius-only arm loses almost the entire improvement over cosine on the standard CLIP matrix: its deficit is 2.53–10.62 points. Thus the large gain over cosine is principally a covariance effect. Radius treatment provides a smaller but distinct gain: replacing it by unit prototypes loses accuracy in thirteen of sixteen settings and ties in three, with the largest losses (2.95 and 2.99 points) on MAE/CUB and IN1K-RN50/CIFAR. A unit prototype discards the norm of the sample mean, although that norm records how concentrated the class is in the representation. A raw five-shot norm retains this signal but is biased upward by the sampling term $1/K$; the corrected arm subtracts that term before nonnegative clipping. The cross-family losses therefore show that radius is useful as a concentration statistic in some representations, not as an independent global logit scale. This interaction explains why the normalized-only method appeared adequate on several CLIP rows but did not transfer as a complete design.

The full-covariance column shows why shrinkage cannot be fixed globally. It ties the selected metric in five settings, is better by at most .43 point in four, and is worse by as much as 2.42 points. Full covariance is most costly on IN1K-RN50/miniImageNet, whereas the selector retains it on several CIFAR rows. Even within one dataset the preferred endpoint changes with the encoder, ruling out a dataset-only explanation. In the eigenspace view, small covariance directions receive large inverse weights: this can expose discriminative structure, but it also magnifies the component of five-shot mean error in those directions. The between-class spectrum and the mean-error spectrum both change with the representation. The selector identifies which effect wins on base episodes; it does not assert that shrinkage itself is always beneficial.

C.3. Selection transfer and auxiliary choices

The retrospective oracle in Table 16 evaluates all 21 candidates with target labels only after the reported candidate is fixed. It measures selector regret rather than choosing a method.

The fixed candidate is the target oracle on six settings, within .22 point on another eight, and never more than .45 point behind. Its exact identity changes in 19 of 64 leave-one-fold-out trials, but the prototype mode is retained in 93.8% and every resulting score remains within .25 point. This distinction is important: the base objective can be flat among nearby shrinkage values, so exchanging adjacent λ values need not constitute a meaningful instability. The retained scores show that the decision geometry transfers even when the discrete argmax over the grid changes. The stable object is therefore audited performance on a flat grid, not parameter identity; we do not claim exact recovery of a unique λ.

Table 18 asks whether the result instead depends on the fixed ridge or on updating covariance from five novel examples.

Table 13. Adaptation-budget comparison on frozen CLIP ViT-B/16. Every row reads the identical frozen features, BiMC sessions, and twenty matched support draws, holding the representation and evaluation protocol fixed while exposing each method's budget. Values are session-average accuracy (mean±std) with final-session Base/Novel, and Δ is the paired ShrinkSel minus that row with a 95% bootstrap interval. No row reads query labels, a class marginal, class names, or CLIP text. The reported adapter mixing weights and replay step count are selected on class-disjoint base episodes, and the fixed choices are evaluated on held-out base classes; optimizer constants are fixed and saved. The adapter rows optimize once on base-session training images; the joint probe revisits those images at every session.

Method	Budget	Session avg.	Final Base	Final Novel	Δ ShrinkSel [95%]
CUB-200					
Cosine-NCM	Support-only; training-free	73.09±0.29	79.88	57.32	+3.41 [+3.33, +3.49]
ShrinkSel (ours)	Support-only; training-free	76.50±0.33	83.41	62.28	—
CLIP-Adapter (image-only)	base-session optimization	71.59±0.27	80.69	50.12	+4.91 [+4.75, +5.08]
ADAM/APER-style concatenation	base-session optimization	73.73±0.28	81.50	55.49	+2.78 [+2.62, +2.93]
Joint linear probe (base replay)	per-session optimization + replay	72.23±0.17	84.20	48.31	+4.27 [+4.15, +4.39]
CIFAR-100					
Cosine-NCM	Support-only; training-free	67.88±0.37	73.33	45.53	+6.19 [+6.09, +6.29]
ShrinkSel (ours)	Support-only; training-free	74.07±0.37	81.22	49.41	—
CLIP-Adapter (image-only)	base-session optimization	72.61±0.39	82.97	35.74	+1.46 [+1.30, +1.62]
ADAM/APER-style concatenation	base-session optimization	74.14±0.32	83.52	40.52	-0.07 [-0.20, +0.06]
Joint linear probe (base replay)	per-session optimization + replay	64.73±0.01	83.94	0.02	+9.34 [+9.17, +9.49]
miniImageNet					
Cosine-NCM	Support-only; training-free	89.00±0.21	91.60	79.76	+2.48 [+2.42, +2.54]
ShrinkSel (ours)	Support-only; training-free	91.48±0.23	94.98	79.96	—
CLIP-Adapter (image-only)	base-session optimization	85.94±0.27	95.50	53.51	+5.54 [+5.41, +5.69]
ADAM/APER-style concatenation	base-session optimization	89.17±0.24	95.75	67.61	+2.31 [+2.20, +2.42]
Joint linear probe (base replay)	per-session optimization + replay	79.32±0.31	96.08	29.06	+12.16 [+11.98, +12.35]

Table 14. Class-balanced joint linear-probe control on frozen CLIP ViT-B/16 features. The probe revisits complete base training features and accumulated labeled novel supports at every session, but fits no query. Δ is ShrinkSel minus the probe over 20 paired supports.

	Session average			ShrinkSel final		Balanced probe final	
Dataset	ShrinkSel	Probe	Δ [95% CI]	Base	Novel	Base	Novel
CUB-200	76.50	74.97	+1.53 [+1.40, +1.66]	83.41	62.28	83.04	57.29
CIFAR-100	74.07	74.62	-0.55 [-0.63, -0.48]	81.22	49.41	82.82	45.78
miniImageNet	91.48	90.89	+0.59 [+0.51, +0.68]	94.98	79.96	95.74	75.35

The target-best ridge equals the reported $\gamma = 1$ in twelve of sixteen settings. The two visible retrospective exceptions are CLIP/CUB (+.39) and Aircraft (+.81), but using those target labels would violate the protocol. Adding novel supports to the pooled covariance changes accuracy by at most .39 point and is negative in several rows. Each five-shot class contributes only four within-class residual degrees of freedom, while the base covariance aggregates complete classes. Novel supports can therefore move a class mean substantially but add little stable second-order information; their noisy covariance contribution can also perturb directions shared by every class. Freezing the base estimate is consequently a robustness choice, not a claim that novel covariance is mathematically irrelevant.

Figure 4 explains the dataset dependence of the normalized-only ablation. CIFAR-100 is nearly flat from full covariance through moderate shrinkage and then drops sharply toward cosine, indicating that many covariance directions remain useful. CUB-200 and miniImageNet instead peak at an interior value: full covariance retains discriminative anisotropy but overweights a subset of noisy directions, while the cosine endpoint discards all anisotropy. The broad plateaus also explain the leave-one-fold result above: a selector can change its discrete λ without materially changing accuracy. Because the figure fixes unit prototypes, its optimum need not match the joint radius–geometry selection in Table 9.

D. Complete Support-Only Results

Tables 19–21 contain every support-only row, including Euclidean NCM, normalized full covariance, diagonal LDA, base-selected TEEN/ULA, and RanPAC without projection. The main paper reports only the ViT-B/16 display set; the complete ViT-B/32 and RN50 matrices are retained here.

Table 15. Factorial support-only component ablation over twenty matched supports. Entries are ShrinkSel minus the named arm in session-average accuracy points.

Encoder	Dataset	Cosine	Shared LDA	Geometry only	Radius only	Full covariance
CLIP ViT-B/16	CUB-200	+3.41	+0.27	+0.08	+3.53	+0.20
CLIP ViT-B/16	CIFAR-100	+6.17	+0.06	+0.23	+6.23	+0.00
CLIP ViT-B/16	miniImageNet	+2.48	+0.29	+0.49	+2.53	+0.00
CLIP ViT-B/16	FGVC-Aircraft	+6.67	+0.00	+0.18	+6.69	+0.00
CLIP ViT-B/32	CUB-200	+5.41	+0.49	+0.28	+5.53	+0.40
CLIP ViT-B/32	CIFAR-100	+6.04	+0.05	+0.39	+6.00	-0.03
CLIP ViT-B/32	miniImageNet	+2.72	+0.30	+0.36	+2.78	-0.03
CLIP RN50	CUB-200	+5.84	+1.17	+0.00	+5.84	+1.46
CLIP RN50	CIFAR-100	+10.71	+0.04	+0.81	+10.62	+0.00
CLIP RN50	miniImageNet	+4.24	+0.66	+0.70	+4.31	+0.13
MAE-ft ViT-B/16	CUB-200	+6.31	+0.90	+2.95	+6.48	+0.00
MAE-ft ViT-B/16	CIFAR-100	+1.76	+0.78	+0.96	+2.15	-0.43
MAE-ft ViT-B/16	miniImageNet	+0.00	+0.65	+0.00	+0.00	+0.21
IN1K RN50	CUB-200	+5.39	+0.74	+1.50	+5.62	+0.10
IN1K RN50	CIFAR-100	+6.48	+0.50	+2.99	+6.62	-0.12
IN1K RN50	miniImageNet	+0.29	+1.52	+0.00	+0.29	+2.42

Table 16. Retrospective target oracle over the selector's 3-by-7 candidate grid. Target labels are used only for this analysis and never replace the base-selected candidate.

Encoder	Dataset	Selected mode	λ	Rank/21	Oracle regret
CLIP ViT-B/16	CUB-200	corrected	0.75	5	0.34
CLIP ViT-B/16	CIFAR-100	corrected	0.0	4	0.12
CLIP ViT-B/16	miniImageNet	corrected	0.0	4	0.11
CLIP ViT-B/16	FGVC-Aircraft	raw	0.0	8	0.45
CLIP ViT-B/32	CUB-200	corrected	0.5	1	0.00
CLIP ViT-B/32	CIFAR-100	corrected	0.1	2	0.01
CLIP ViT-B/32	miniImageNet	corrected	0.25	2	0.09
CLIP RN50	CUB-200	unit	0.75	1	0.00
CLIP RN50	CIFAR-100	corrected	0.0	2	0.00
CLIP RN50	miniImageNet	corrected	0.5	1	0.00
MAE-ft ViT-B/16	CUB-200	corrected	0.0	3	0.21
MAE-ft ViT-B/16	CIFAR-100	corrected	0.5	4	0.10
MAE-ft ViT-B/16	miniImageNet	unit	1.0	3	0.02
IN1K RN50	CUB-200	corrected	0.75	1	0.00
IN1K RN50	CIFAR-100	corrected	0.25	2	0.22
IN1K RN50	miniImageNet	unit	0.9	1	0.00

Every row shares the same features, supports, sessions, and query manifests. Their rankings support the same bounded claim: ShrinkSel improves shared LDA on the nine core pairs, although several gains are deliberately small because the robust rule stays near its LDA fallback.

The main paper gives selected ViT-B/16 session curves. Projected RanPAC begins strongly on the base session but degrades rapidly as five-shot classes accumulate, particularly on CIFAR-100 and miniImageNet. Shared LDA and ShrinkSel remain close throughout, with the selected ShrinkSel row slightly higher in session-average accuracy.

D.1. Baseline implementation provenance

All functions are query-local and operate on identical cached features. The following details address differences that are easy to hide under method names.

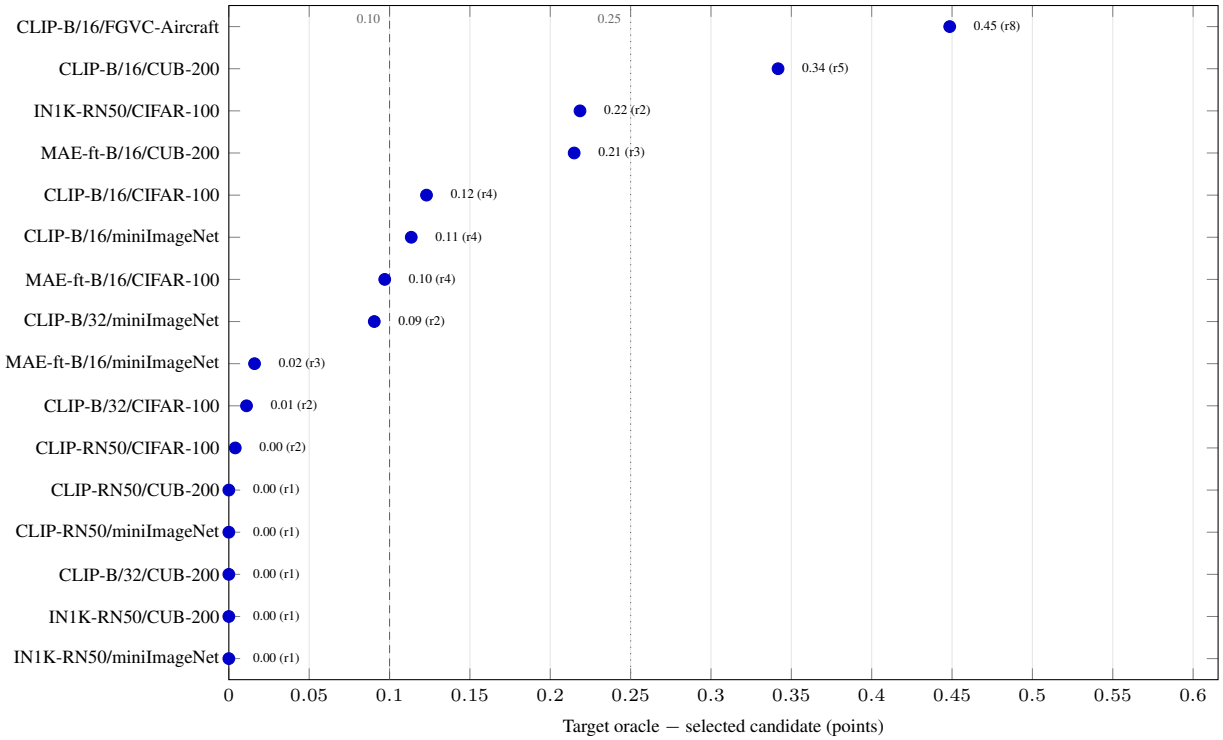


Figure 3. Target-oracle regret of the fixed support-only selector on every evaluated backbone–dataset setting. The oracle evaluates all 21 candidates on target labels only after selection; it is a diagnostic and never selects a reported method. Parentheses give the selected candidate's target rank.

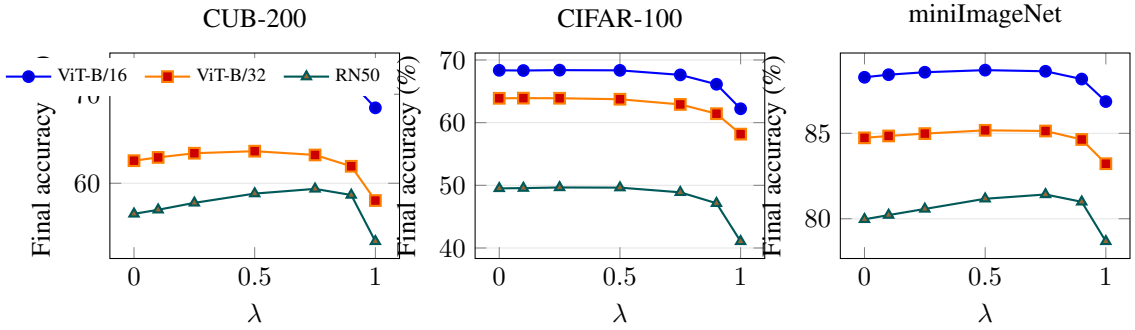


Figure 4. Retrospective target sensitivity of the *normalized-only* shrinkage path on the nine core backbone–dataset pairs. Values are final-session overall accuracy averaged over twenty support seeds; $\lambda = 0$ retains the most covariance shape and $\lambda = 1$ is the cosine endpoint. These curves diagnose why covariance selection alone could not recover raw-mean LDA; they are not inputs to the joint selector.

Cosine, Euclidean, and LDA. Cosine uses normalized arithmetic means. Euclidean and shared/diagonal LDA use raw arithmetic means. Shared LDA uses the full base covariance with ridge $\gamma\delta I$; diagonal LDA retains only its coordinate variances. Novel supports update means only.

TEEN and ULA. TEEN [12] forms a softmax-weighted combination of normalized base prototypes and interpolates it with each normalized novel prototype. We report both released defaults and class-disjoint base-selected parameters. ULA [13] mirrors released Case A: when a query's top class is base, runner-up is novel, and their margin is small, it shifts the two logits using a base-entropy gate. It never aggregates queries.

FeCAM and HyCal. FeCAM [3] uses its released ViT covariance-shrink/correlation settings. We omit its fractional Tukey transform because normalized CLIP coordinates can be negative and the released fractional power would be undefined. HyCal [4] is reconstructed from its

Table 17. Base-only leave-one-fold-out selector stability over all 16 settings (64 omissions). Exact identity is stricter than practical score stability.

Criterion	Result
Exact candidate retained	45/64 (70.3%)
Prototype mode retained	93.8%
Selected score within 0.10 point	96.9%
Selected score within 0.25 point	100.0%
Selected candidate infeasible	0

paper equations because no official code was linked when the experiments were specified. At its standard five-shot parameters $(\alpha, \beta) = (1, 0)$, we interpret the otherwise singular $\beta = 0$ sigmoid as the paper's intended hard threshold; the Mahalanobis weight is then one for every five-shot class. Translating a query and its class prototype by the same text vector cancels in the difference, and covariance is translation-invariant; the evaluated limiting row is therefore exactly image-only. The published rule places a nonnegative Mahalanobis distance in an argmax without specifying a released implementation. We use $-d$, the standard nearest-Mahalanobis interpretation; consequently this is an equation-level reconstruction rather than an asserted official-code reproduction.

RanPAC and RanDumb. Projected RanPAC [7] uses a fixed seed-1993 Gaussian 10k projection, ReLU, accumulated normal equations, and the official ridge grid $10^{-8}, \ldots, 10^{8}$. The support update uses Woodbury algebra; unit tests compare it with a direct solve. The official ridge objective is unchanged, but its numerical solver is adaptive. If the number of base examples is smaller than the 10k projected dimension, we diagonalize the float64 sample-space Gram matrix and solve in its nonzero right-singular subspace; otherwise we use the primal Cholesky solve. These are algebraically equivalent forms of $(H^\top H + \lambda I)W = H^\top Y$. Every evaluated run must satisfy a scale-free normal-equation residual tolerance before its accuracy is accepted.

Every accepted projected-RanPAC result must pass the residual check. The MAE-ft/CUB row is 67.21 session-average with residual 3.71×10^{-12}; the remaining five cross-family residuals also pass the stated tolerance. RanDumb [9] uses 10k random Fourier features with seed 1993, $\gamma = 1/(D \operatorname{Var}[z])$ estimated from construction classes, and cosine NCM in the transformed space. Random weights, offsets, and selected parameters are recorded in the release.

E. FGVC-Aircraft Extension and Data Provenance

The Aircraft [6] extension uses the official 2013b archive. Before extraction, we verify its pinned byte length (2,753,340,328), reject links, validate archive members, and check the complete train+validation/test manifests. The validated copy contains 10,000 task images, 6,667 train+validation examples and 3,333 test examples. The fixed train/test manifests are included in the released artifacts. The published big-start structure does not supply a universal Aircraft class index list, so we use the seed-1993 order of the evaluated FSCIL-calibration configuration and serialize the full permutation.

Aircraft satisfies the diagnostic eligibility rule used for this extension: the lower 95% endpoint of r exceeds .5, the full-mean replacement term exceeds two points, and $A_n^{\text{full}} \geq 20\%$. The five-draw diagnostic obtains $A_n^K = 25.81$, $A_n^{\text{full}} = 49.21$, a full-mean replacement term of 23.40, and $r = .95$ [.87, 1.03]. The held-out base safety criterion also passes. The reported method matrix uses twenty matched support draws and appears in Table 19.

F. Resource accounting and chunk scaling

Resource experiments run on an NVIDIA RTX 3090 (24 GiB), PyTorch 1.13.1+cu117, CUDA 11.7, and Python 3.8.20. We time three distinct operations: base-head construction, the final-session labeled-support update, and synchronized query scoring. Score time is the median of five repetitions after two warmups except chunk 1, which uses three repetitions after one warmup. Feature extraction and disk I/O are excluded. Persistent state counts method tensors; peak is CUDA memory allocated during scoring, including resident evaluation tensors.

Table 26 reports the complete ShrinkSel chunk sweep. Chunk 1 is the relevant upper bound for unbatched kernel-launch overhead: about 29–30 μs/query. Full-tensor throughput is 0.069–0.070 μs/query. The accuracy formula is unchanged by this computational chunk; the two near ties described in Sec. A occur in other methods.

For ShrinkSel, one base covariance/eigensystem and all prototypes require $O(D^2+CD)$ state. FeCAM stores a class-specific $D \times D$ precision and therefore scales as $O(CD^2)$. HyCal uses an exact low-rank representation, but its per-class SVDs still dominate support-update time. These measured differences explain the large state and peak-memory gap in Table 25.

G. Query-Assisted Evaluation

G.1. Identity-disjoint calibration

QPR is introduced in this work. Its reported parameters come from a base-only calibration in which anchor and pseudo-novel semantic identities are disjoint by construction. Construction identities provide anchor prototypes and held-out anchor queries. Separate selection identities provide each five-shot pseudo-novel support and its queries. The prototype treatment already selected for the parent ShrinkSel head

Table 18. Auxiliary support-only diagnostics. Best γ is the target-diagnostic ridge winner; its gain never changes the reported $\gamma = 1$ row. The final column adds only labeled novel supports to covariance estimation.

Encoder	Dataset	Best γ	Best minus reported	Novel-support covariance minus base-only [95% CI]
CLIP ViT-B/16	CIFAR-100	1.0	+0.00	+0.02 [+0.01, +0.02]
CLIP ViT-B/16	CUB-200	0.1	+0.39	+0.05 [-0.00, +0.09]
CLIP ViT-B/16	miniImageNet	1.0	+0.00	+0.02 [+0.01, +0.02]
MAE-ft ViT-B/16	CIFAR-100	0.1	+0.10	+0.03 [+0.02, +0.03]
MAE-ft ViT-B/16	CUB-200	1.0	+0.00	+0.39 [+0.34, +0.43]
MAE-ft ViT-B/16	miniImageNet	0.01	+0.00	+0.00 [+0.00, +0.00]
CLIP RN50	CIFAR-100	1.0	+0.00	-0.02 [-0.03, -0.01]
CLIP RN50	CUB-200	1.0	+0.00	+0.11 [+0.06, +0.15]
CLIP RN50	miniImageNet	1.0	+0.00	+0.01 [+0.00, +0.02]
CLIP ViT-B/16	FGVC-Aircraft	0.1	+0.81	+0.10 [+0.06, +0.14]
CLIP ViT-B/32	CIFAR-100	1.0	+0.00	-0.00 [-0.01, +0.01]
CLIP ViT-B/32	CUB-200	1.0	+0.00	-0.02 [-0.06, +0.01]
CLIP ViT-B/32	miniImageNet	1.0	+0.00	+0.01 [+0.00, +0.01]
IN1K RN50	CIFAR-100	1.0	+0.00	-0.01 [-0.02, +0.00]
IN1K RN50	CUB-200	1.0	+0.00	+0.20 [+0.15, +0.25]
IN1K RN50	miniImageNet	1.0	+0.00	+0.00 [-0.00, +0.01]

Table 19. Complete support-only session-average matrix for CLIP ViT-B/16 (mean±std over 20 matched support seeds). Bold marks the best complete row per dataset. †HyCal denotes only our algebraically image-only hard-threshold limit of its published five-shot setting.

Method	CUB-200	CIFAR-100	miniImageNet	FGVC-Aircraft
Cosine-NCM	73.09±0.29	67.88±0.37	89.00±0.21	46.24±0.33
Euclidean-NCM	72.97±0.28	67.80±0.37	88.81±0.20	46.22±0.35
Normalized covariance ($\lambda = 0$)	75.99±0.44	73.84±0.39	90.99±0.27	52.73±0.33
Shared-covariance LDA	76.24±0.38	74.01±0.38	91.19±0.22	**52.91**±0.26
Diagonal LDA	74.63±0.35	70.31±0.36	90.51±0.22	47.20±0.33
ShrinkSel	**76.50**±0.33	**74.07**±0.37	**91.48**±0.23	**52.91**±0.26
TEEN (released)	71.95±0.37	67.80±0.38	89.21±0.20	45.73±0.37
TEEN (base-selected)	72.62±0.32	68.08±0.35	89.08±0.21	46.53±0.45
ULA (released)	73.15±0.29	68.46±0.35	89.31±0.26	46.50±0.48
ULA (base-selected)	72.02±0.27	67.46±0.46	89.43±0.32	45.93±0.43
FeCAM	60.00±0.27	60.47±0.09	71.94±0.27	45.74±0.33
HyCal image-only reduction†	61.54±0.28	64.07±0.05	71.40±0.03	45.38±0.17
RanPAC (no projection)	57.01±0.10	64.42±0.01	73.54±0.01	47.21±0.18
RanPAC (10k projection)	70.44±0.33	67.54±0.14	81.53±0.32	49.70±0.20
RanDumb (RBF-NCM)	72.80±0.30	67.51±0.37	88.12±0.21	46.63±0.32

is applied to both groups. Every serialized episode records zero identity-overlap violations, and target arrays remain unavailable to selection and held-out base safety checking.

The one-increment tuples $(\alpha, \tau_q, m_{\min}, \kappa)$ are $(.5, .3, 5, .3)$ on CUB-200, $(.5, .3, 5, 1)$ on CIFAR-100, and $(.75, .3, 5, 1)$ on miniImageNet. All pass the held-out base safety gates. The audit class-balanced gains over the parent ShrinkSel head are $.48$, $.43$, and $.27$ point, while anchor changes are $-.37$, $-.09$, and $-.07$ point, respectively. The update therefore clears both the $.10$-point combined non-regression gate and the $.50$-point anchor-loss gate before target evaluation.

G.2. Cumulative-competition stress test

The primary selectors request one public class increment per episode. We rerun both ShrinkSel and QPR calibration with requested pseudo-novel counts cycling through the public cumulative schedule. The class-disjoint pseudo folds contain at most 20 identities, so requested later-session counts are clipped: 80% of CUB episodes and 75% of CIFAR/miniImageNet episodes saturate. Requested and realized histograms are stored in the machine-readable artifacts. This is therefore a bounded competition stress test, not an exact emulation of 40–100 target novel classes.

The cumulative ShrinkSel calibration selects exactly the

Table 20. Complete support-only session-average matrix for CLIP ViT-B/32 (mean±std over 20 matched support seeds). Bold marks the best complete row per dataset. †HyCal denotes only our algebraically image-only hard-threshold limit of its published five-shot setting.

Method	CUB-200	CIFAR-100	miniImageNet
Cosine-NCM	63.71±0.37	64.43±0.35	85.34±0.20
Euclidean-NCM	63.56±0.37	64.43±0.35	85.13±0.22
Normalized covariance ($\lambda = 0$)	68.20±0.42	70.08±0.39	87.62±0.23
Shared-covariance LDA	68.63±0.44	70.42±0.38	87.76±0.19
Diagonal LDA	65.77±0.30	66.65±0.37	87.13±0.20
ShrinkSel	**69.13**±0.39	**70.47**±0.37	**88.06**±0.20
TEEN (released)	62.81±0.34	64.36±0.33	85.72±0.23
TEEN (base-selected)	63.48±0.35	64.66±0.36	85.58±0.23
ULA (released)	63.97±0.40	64.99±0.41	85.70±0.21
ULA (base-selected)	62.87±0.41	64.33±0.45	85.86±0.29
FeCAM	52.19±0.26	58.60±0.09	69.64±0.28
HyCal image-only reduction†	53.49±0.21	62.29±0.04	69.84±0.01
RanPAC (no projection)	52.77±0.09	62.25±0.01	71.64±0.02
RanPAC (10k projection)	61.63±0.33	66.23±0.14	78.73±0.26
RanDumb (RBF-NCM)	63.57±0.36	64.51±0.36	84.39±0.20

Table 21. Complete support-only session-average matrix for CLIP RN50 (mean±std over 20 matched support seeds). Bold marks the best complete row per dataset. †HyCal denotes only our algebraically image-only hard-threshold limit of its published five-shot setting.

Method	CUB-200	CIFAR-100	miniImageNet
Cosine-NCM	59.69±0.39	47.61±0.32	81.12±0.28
Euclidean-NCM	59.59±0.37	47.71±0.32	80.86±0.28
Normalized covariance ($\lambda = 0$)	64.07±0.36	57.51±0.41	84.02±0.36
Shared-covariance LDA	64.36±0.33	58.28±0.34	84.71±0.29
Diagonal LDA	64.09±0.35	52.66±0.32	84.21±0.24
ShrinkSel	**65.53**±0.37	58.32±0.35	**85.36**±0.25
TEEN (released)	58.73±0.41	47.20±0.26	81.31±0.28
TEEN (base-selected)	59.39±0.34	47.43±0.28	81.22±0.35
ULA (released)	59.94±0.46	47.59±0.37	81.48±0.30
ULA (base-selected)	59.05±0.48	46.69±0.45	81.67±0.35
FeCAM	42.35±0.18	41.03±0.01	61.57±0.14
HyCal image-only reduction†	50.45±0.13	54.79±0.01	68.16±0.01
RanPAC (no projection)	49.69±0.13	55.24±0.02	70.90±0.04
RanPAC (10k projection)	57.82±0.30	**58.77**±0.09	75.80±0.26
RanDumb (RBF-NCM)	59.51±0.37	47.76±0.31	80.03±0.26

primary head on all three datasets and passes every held-out base safety check. The QPR rows report an independent cumulative calibration with the same grid, objective, identity separation, and safety rules; no target result chooses between the one-increment and cumulative tuples. Cumulative QPR selects $(.5, .9, 5, .3)$ on CUB-200, retains $(.5, .3, 5, 1)$ on CIFAR-100, and selects $(.5, .3, 5, 1)$ on miniImageNet. Its same-cohort gains remain 1.46, 2.03, and 1.11 points, while exact LOO retains 1.64, 2.03, and 1.10 points. Thus the qualitative QPR result survives the stronger competition surrogate, although CUB-200 and miniImageNet do not retain the same tuple and the clipping boundary prevents an exact late-session claim.

G.3. Exact query-self-exclusion

Ordinary QPR is same-cohort batch transduction: the unlabeled batch constructs the update and is then rescored. To isolate query self-inclusion, the leave-one-out (LOO) variant removes query i from the update used to classify i. Because admission is a query-local function of fixed pre-update scores, exact LOO requires changing only the score against the admitted novel class. The implementation subtracts the query's posterior-weighted feature from that class aggregate; when the remaining group falls below $m_{\min}$, it restores the support-only prototype for that query. Synthetic tests compare every vectorized score with a literal rerun after removing the query.

Under the primary calibration, LOO retains gains of 1.84, 2.03, and 1.19 session-average points over ShrinkSel. Its changes from same-cohort QPR are $+.22$, $-.01$, and $-.02$ point. Thus self-inclusion is not needed for the reported improvement. LOO still consumes the other queries in the cohort, so it establishes a held-out-query property rather than support-only or streaming inference.

As a cross-machine check, we replayed the first support draw at the final session on a second host using CPU execution and the identical feature, protocol, selection, and audit records. All ShrinkSel, same-cohort QPR, and LOO Full/Base/Novel accuracies agree with the primary GPU artifacts to within 6×10^{-6} percentage point. This is an implementation reproducibility check, not an additional statistical sample.

The final-session decomposition shows the remaining trade-off directly: novel accuracy rises substantially while base accuracy falls. QPR changes only novel prototypes, but a moved novel boundary can capture a base query. Query self-exclusion does not remove that mechanism.

G.4. Components of the operator

The selected operator moves the mean novel-prototype direction toward the complete- training mean on all three datasets: $9.62^\circ \to 6.79^\circ$ on CUB-200, $10.03^\circ \to 6.80^\circ$ on CIFAR-100, and $16.25^\circ \to 6.55^\circ$ on miniImageNet. Radius preservation is the only single removal with a material loss on every dataset ($-.45$, -1.62, and -1.90 points). Removing all conservative controls is worse by 2.61, 1.56, and 3.05 points. The mass and minimum-count gates are nearly inactive on these complete target cohorts, while uniform weights, the predicted-novel gate, and a full directional step have small, dataset-dependent effects. We therefore interpret the ablation as support for controlled direction-only rectification, not as proof that every individual gate is necessary on every target.

The component variants are post-selection diagnostics on target data; they do not replace the fixed base-selected tuple. Complete novel-class means and query labels are read only after prediction to measure angles, repairs, and

Table 22. Cross-family support-only session-average accuracy (mean±std over 20 matched support draws). FeCAM is omitted at 2048 dimensions by the fixed $D \leq 1024$ feasibility rule: its class-specific dense implementation scales as $O(CD^2)$ state, $O(CD^3)$ factorization, and $O(QCD^2)$ scoring; for CUB, state alone is 3.12 GiB.

Backbone	Method	CUB-200	CIFAR-100	*mini*ImageNet
MAE-ft ViT-B/16	Cosine-NCM	63.24±0.35	71.97±0.29	98.76±0.06
	TEEN (released)	62.77±0.35	72.92±0.33	98.53±0.06
	ULA (released)	63.56±0.37	72.38±0.31	**98.77**±0.04
	FeCAM	60.13±0.21	65.01±0.08	86.26±0.31
	HyCal image-only	59.67±0.16	64.75±0.04	79.45±0.21
	RanPAC (10k projection)	67.21±0.38	68.36±0.11	94.21±0.12
	Shared-covariance LDA	68.66±0.27	72.94±0.26	98.12±0.07
	ShrinkSel	**69.55**±0.25	**73.72**±0.29	98.76±0.06
IN1K RN50	Cosine-NCM	53.96±0.38	56.95±0.25	95.48±0.20
	TEEN (released)	53.61±0.43	57.59±0.35	95.16±0.21
	ULA (released)	54.36±0.41	57.41±0.27	95.54±0.20
	HyCal image-only	51.44±0.20	59.68±0.01	77.80±0.23
	RanPAC (10k projection)	58.46±0.32	62.69±0.06	90.06±0.18
	Shared-covariance LDA	58.61±0.33	62.94±0.27	94.25±0.26
	ShrinkSel	**59.35**±0.37	**63.43**±0.28	**95.77**±0.19

Table 23. Implementation provenance. Comparator rows identify upstream sources inspected before the clean-room matched implementations; QPR is introduced in this work. Full revisions and licensing notes are preserved in the code release.

Method	Upstream/reconstruction source
RanPAC	Official RanPAC repository
RanDumb	Official RanDumb repository
TEEN	Official TEEN repository
ULA	Official ULA repository
FeCAM	Official FeCAM repository
HyCal	CVPR 2026 paper equations (arXiv 2604.15678v1)
ProtoLP	CVPR 2023 Eqs. 12–18 + repository
TIM	Official TIM repository
LaplacianShot full/core	Official repository
MAP-OT (PT-MAP core)	Official repository
QPR	This work

Table 24. Support-only FGVC-Aircraft accuracy with frozen CLIP ViT-B/16 (mean±std over twenty matched support draws).

Method	Session avg.	Final session
Cosine-NCM	46.24±0.33	38.97±0.56
TEEN (released)	45.73±0.37	37.92±0.49
RanDumb (RBF-NCM)	46.63±0.32	38.84±0.48
RanPAC (10k projection)	49.70±0.20	38.10±0.30
Shared-covariance LDA	**52.91**±0.26	**43.20**±0.47
ShrinkSel	**52.91**±0.26	**43.20**±0.47

harms. Accordingly, a target variant that happens to improve a dataset is evidence about mechanism, not a second hyperparameter-selection stage.

Table 25. Full-tensor CIFAR-100/ViT-B/16 resource comparison. Score time is batched amortized throughput, not synchronous single-image latency.

Method	Build (ms)	Update (ms)	Batched μs/query	State (MiB)	Peak (MiB)
Cosine-NCM	2.4	3.7	0.02	0.20	534.8
TEEN (released)	2.4	3.9	0.02	0.20	534.8
Shared-covariance LDA	546.9	3.7	0.05	1.20	567.3
ShrinkSel	546.9	3.7	0.07	1.20	567.3
FeCAM	1124.1	609.9	4.65	100.20	6393.9
HyCal image-only reduction†	1461.9	19.8	4.75	59.29	569.4

G.5. Cohort, shot, and selector robustness

The selected tuple is held fixed while the target cohort changes from the complete set to balanced subsets, random

Table 26. ShrinkSel query-chunk scaling on CLIP ViT-B/16. "Queries" is the complete final-session evaluation set; Chunk is the number scored per matrix block.

Chunk	Queries	Median score (ms)	Batched μs/query	Peak (MiB)
CUB-200				
1	5794	175.062	30.214	83.5
16	5794	11.291	1.949	492.4
64	5794	3.327	0.574	492.4
256	5794	1.039	0.179	492.5
1024	5794	0.675	0.116	492.9
full	5794	0.399	0.069	506.8
CIFAR-100				
1	10000	302.547	30.255	275.1
16	10000	19.242	1.924	535.1
64	10000	5.580	0.558	535.0
256	10000	1.437	0.144	535.0
1024	10000	0.672	0.067	535.3
full	10000	0.702	0.070	567.3
miniImageNet				
1	10000	291.776	29.178	275.1
16	10000	19.747	1.975	535.1
64	10000	5.738	0.574	535.0
256	10000	1.428	0.143	535.0
1024	10000	0.673	0.067	535.3
full	10000	0.698	0.070	567.3

Table 27. Cumulative-competition calibration. Heads are prototype treatment / λ; Δ is the paired all-session target change from the primary one-increment-selected ShrinkSel, with a 95% bootstrap interval.

Dataset	1-inc.	Cumul.	Δ [95% CI]
CUB-200	corr. / 0.75	corr. / 0.75	+0.00 [+0.00,+0.00]
CIFAR-100	corr. / 0	corr. / 0	+0.00 [+0.00,+0.00]
miniImageNet	corr. / 0	corr. / 0	+0.00 [+0.00,+0.00]

batches, causal windows, and deterministic 10:1 or 100:1 long-tailed subsets. Random and causal constructions use five feature-independent order seeds crossed with all twenty supports. Separately, the five-shot-selected tuple is applied without reselection at $K \in \{1, 2, 5, 10, 20\}$. These target diagnostics test a frozen operator; none chooses a preferred mode, shot count, or tuple.

The interaction check crosses the same five support counts with balanced cohorts of $1, 2, 5, 10$, or 20 queries per seen class at the final session. It uses the same twenty nested supports and cohort seed offset zero. The $K = 5$ row is replay-identical to the validated cohort artifact; the other twelve dataset–shot artifacts were run with the unchanged operator. Each cell in the main heatmap is accompanied in the analysis JSON by its twenty paired differences and 10,000-resample interval.

Complete-cohort gains are positive on all three evaluated datasets, but QPR is not uniformly helpful as the cohort shrinks. Balanced cohorts with one or two queries per class yield essentially no gain, and five queries can be adverse. All balanced ten- and twenty-query rows are positive. Random or causal groups of 512 are adverse on CUB-200 but positive on the other two datasets, whereas both deterministic long-tail modes remain positive on all three. These outcomes rule out a cohort-size- or composition-free improvement claim.

The shot sweep has the complementary pattern. Gains peak at one or two shots, remain positive through ten, and approach zero at twenty. At $K = 20$, the CIFAR-100 and miniImageNet intervals include zero; only CUB-200 excludes zero, and its .07-point effect is practically tiny. This is consistent with QPR repairing scarce estimates rather than supplying a fixed improvement independent of support size.

Architecture sensitivity is checked on CLIP ViT-B/32 and RN50 in addition to the primary ViT-B/16 representation. Each added backbone receives a separate class-disjoint base selection and safety check, followed by the fixed full-cohort target run. This tests sensitivity within the CLIP family and does not establish transfer to an unrelated representation family.

All six added target runs improve their matched ShrinkSel parent by 1.34–2.22 session-average points with paired intervals above zero. As on ViT-B/16, every final result loses base accuracy and gains substantially more novel accuracy. This supports architecture robustness within CLIP; the tested family does not support a broader representation-family claim.

Selector stability is evaluated without target arrays by repeating the complete calibration on five deterministic, class-disjoint construction/selection/safety re-splits. We retain every selected tuple, safety outcome, and audit regret to the best audit-safe grid point; target accuracy cannot select a preferred partition.

All fifteen re-split audits pass, and target-free regret to the best audit-safe point is at most $.05$ point. Parameter identity is less stable: CUB-200 selects a different tuple in every re-split, while CIFAR-100 and miniImageNet select two and three unique tuples. The appropriate conclusion is that several grid points are audit-equivalent, not that the same tuple is recovered from every base partition.

G.6. Matched transductive comparators

The QPR predictions use the identical frozen features, support seeds, sessions, and complete all-seen query cohorts as the immutable comparator runs. Comparator predictions are not rerun or altered. The ProtoLP support-assignment rows are clamped to one-hot labels as in its equation; only query rows receive soft prototype assignments. LaplacianShot-full implements the released CL2N center-

Table 28. QPR calibration and exact query-self-exclusion. Tuples are $(\alpha, \tau_q, m_{\min}, \kappa)$. Gains are paired all-session accuracy changes with 95% bootstrap intervals.

Dataset	Calibration	Deployed tuple	Same − ShrinkSel	LOO − ShrinkSel	LOO − Same
CUB-200	one increment	(0.5,0.3,5,0.3)	+1.63 [+1.53,+1.71]	+1.84 [+1.73,+1.94]	+0.22 [+0.17,+0.26]
CUB-200	cumulative	(0.5,0.9,5,0.3)	+1.46 [+1.37,+1.54]	+1.64 [+1.53,+1.75]	+0.18 [+0.15,+0.22]
CIFAR-100	one increment	(0.5,0.3,5,1)	+2.03 [+1.93,+2.15]	+2.03 [+1.91,+2.14]	-0.01 [-0.02,-0.00]
CIFAR-100	cumulative	(0.5,0.3,5,1)	+2.03 [+1.93,+2.15]	+2.03 [+1.91,+2.14]	-0.01 [-0.02,-0.00]
miniImageNet	one increment	(0.75,0.3,5,1)	+1.21 [+1.15,+1.28]	+1.19 [+1.12,+1.26]	-0.02 [-0.03,-0.02]
miniImageNet	cumulative	(0.5,0.3,5,1)	+1.11 [+1.05,+1.16]	+1.10 [+1.05,+1.16]	-0.00 [-0.01,+0.00]

Table 29. Final-session Full/Base/Novel accuracy over 20 paired support draws. Parentheses give paired changes from ShrinkSel.

Dataset	ShrinkSel	QPR same cohort	QPR self-excluded
CUB-200	72.72/83.41/62.28	75.25/82.55/68.11 (+2.53/-0.86/+5.83)	75.66/82.59/68.89 (+2.93/-0.82/+6.61)
CIFAR-100	68.50/81.22/49.41	72.05/80.08/59.99 (+3.55/-1.14/+10.58)	72.04/80.10/59.95 (+3.54/-1.12/+10.54)
miniImageNet	88.97/94.98/79.96	91.15/94.32/86.38 (+2.18/-0.66/+6.42)	91.12/94.33/86.30 (+2.15/-0.65/+6.34)

Table 30. Complete final-session component diagnostics for QPR over 20 matched support draws. Each entry is the paired overall-accuracy change from the fixed, base-selected QPR operator with a 95% bootstrap interval. Positive values favor the diagnostic variant. These target-facing removals never select the reported configuration.

Diagnostic variant	Δ [95% CI]
CUB-200	
Uniform posterior weights	-0.162 [-0.208, -0.117]
Remove novel-mass gate ($\tau_q = 0$)	-0.001 [-0.003, +0.000]
Minimum admitted count $m_{\min} = 1$	+0.009 [-0.000, +0.022]
Do not preserve support radius	-0.450 [-0.530, -0.368]
Remove predicted-novel gate	+0.012 [-0.023, +0.050]
Full directional step ($\alpha = 1$)	+0.186 [+0.079, +0.292]
Remove all conservative controls	-2.614 [-2.767, -2.461]
CIFAR-100	
Uniform posterior weights	+0.049 [+0.032, +0.066]
Remove novel-mass gate ($\tau_q = 0$)	+0.001 [+0.000, +0.002]
Minimum admitted count $m_{\min} = 1$	+0.005 [+0.000, +0.012]
Do not preserve support radius	-1.618 [-1.704, -1.535]
Remove predicted-novel gate	+0.044 [+0.027, +0.061]
Full directional step ($\alpha = 1$)	-0.184 [-0.245, -0.122]
Remove all conservative controls	-1.562 [-1.797, -1.329]
miniImageNet	
Uniform posterior weights	+0.031 [+0.020, +0.043]
Remove novel-mass gate ($\tau_q = 0$)	+0.000 [+0.000, +0.000]
Minimum admitted count $m_{\min} = 1$	-0.001 [-0.003, +0.000]
Do not preserve support radius	-1.900 [-1.961, -1.840]
Remove predicted-novel gate	+0.016 [+0.006, +0.024]
Full directional step ($\alpha = 1$)	-0.053 [-0.073, -0.031]
Remove all conservative controls	-3.052 [-3.259, -2.832]

Table 31. QPR cohort robustness at the final session. Values are QPR minus ShrinkSel in points. Random batches and causal windows use five feature-independent order constructions and a crossed order/support bootstrap; other rows use 20 paired supports.

Cohort	Δ Full [95% CI]	Δ Base	Δ Novel
CUB-200			
full	+2.53 [+2.39, +2.66]	-0.86	+5.83
balanced 1	-0.02 [-0.07, 0.00]	0.00	-0.05
balanced 2	-0.13 [-0.24, 0.00]	0.00	-0.25
balanced 5	-0.90 [-1.16, -0.66]	-0.14	-1.67
balanced 10	+1.56 [+1.43, +1.71]	-0.53	+3.67
balanced 20	+2.32 [+2.17, +2.46]	-0.84	+5.45
random 128	0.00 [-0.01, 0.00]	0.00	0.00
random 512	-0.29 [-0.34, -0.24]	-0.05	-0.52
causal 512/128	-0.32 [-0.38, -0.26]	-0.05	-0.58
imbalance 10:1	+1.90 [+1.76, +2.03]	-0.48	+4.06
imbalance 100:1	+1.90 [+1.66, +2.12]	-0.20	+3.71
CIFAR-100			
full	+3.55 [+3.38, +3.73]	-1.14	+10.58
balanced 1	0.00 [0.00, 0.00]	0.00	0.00
balanced 2	0.00 [0.00, 0.00]	0.00	0.00
balanced 5	+0.04 [-0.05, +0.15]	0.00	+0.10
balanced 10	+1.35 [+1.21, +1.49]	-0.30	+3.82
balanced 20	+2.47 [+2.33, +2.62]	-0.66	+7.18
random 128	0.00 [0.00, +0.01]	0.00	+0.01
random 512	+0.33 [+0.29, +0.37]	-0.13	+1.02
causal 512/128	+0.33 [+0.29, +0.37]	-0.13	+1.02
imbalance 10:1	+2.97 [+2.83, +3.11]	-0.74	+9.52
imbalance 100:1	+2.92 [+2.75, +3.10]	-0.42	+8.73
miniImageNet			
full	+2.18 [+2.05, +2.31]	-0.66	+6.42
balanced 1	0.00 [0.00, 0.00]	0.00	0.00
balanced 2	0.00 [0.00, 0.00]	0.00	0.00
balanced 5	-0.07 [-0.14, -0.01]	-0.03	-0.12
balanced 10	+0.71 [+0.61, +0.82]	-0.18	+2.05
balanced 20	+1.46 [+1.30, +1.62]	-0.53	+4.44
random 128	0.00 [0.00, +0.01]	0.00	+0.01
random 512	+0.40 [+0.37, +0.43]	-0.11	+1.17
causal 512/128	+0.39 [+0.36, +0.41]	-0.11	+1.14
imbalance 10:1	+1.51 [+1.35, +1.68]	-0.54	+5.15
imbalance 100:1	+1.60 [+1.43, +1.78]	-0.38	+5.06

ing, query shift/prototype rectification, Euclidean graph, and bound update, whereas the core row applies only the graph operator to the matched normalized features. MAP-OT$_u$

Table 32. QPR shot robustness. The fixed five-shot-selected operator is applied without reselection at every K. Session-average intervals resample 20 paired support draws.

Dataset	K	Δ session avg. [95% CI]	Δ final Full	Δ Base	Δ Novel
CUB-200	1	+2.90 [+2.74, +3.09]	+5.47	-0.12	+10.94
CUB-200	2	+3.75 [+3.59, +3.92]	+6.11	-0.54	+12.61
CUB-200	5	+1.63 [+1.53, +1.71]	+2.53	-0.86	+5.83
CUB-200	10	+0.52 [+0.46, +0.57]	+0.87	-0.73	+2.44
CUB-200	20	+0.07 [+0.03, +0.11]	+0.18	-0.44	+0.79
CIFAR-100	1	+2.06 [+1.85, +2.27]	+3.59	-0.13	+9.16
CIFAR-100	2	+3.17 [+3.03, +3.32]	+5.88	-0.53	+15.50
CIFAR-100	5	+2.03 [+1.92, +2.15]	+3.55	-1.14	+10.58
CIFAR-100	10	+0.75 [+0.69, +0.80]	+1.32	-1.04	+4.86
CIFAR-100	20	+0.04 [0.00, +0.09]	+0.14	-0.67	+1.34
miniImageNet	1	+4.61 [+4.28, +4.93]	+8.17	-0.10	+20.59
miniImageNet	2	+4.65 [+4.42, +4.91]	+8.18	-0.54	+21.28
miniImageNet	5	+1.21 [+1.15, +1.28]	+2.18	-0.66	+6.42
miniImageNet	10	+0.34 [+0.29, +0.39]	+0.64	-0.55	+2.43
miniImageNet	20	+0.01 [-0.02, +0.05]	+0.08	-0.42	+0.83

Table 33. QPR architecture sensitivity at $K = 5$ and with the complete cohort. Each backbone receives a separate class-disjoint base-only selection and audit; target outcomes do not choose among tuples. Values are QPR minus its matched ShrinkSel parent over 20 supports. The comparison is restricted to the CLIP backbone family.

Backbone	Dataset	Δ session avg. [95% CI]	Δ final Full	Δ Base	Δ Novel
ViT-B/16	CUB-200	+1.63 [+1.53, +1.71]	+2.53	-0.86	+5.83
ViT-B/16	CIFAR-100	+2.03 [+1.93, +2.15]	+3.55	-1.14	+10.58
ViT-B/16	miniImageNet	+1.21 [+1.15, +1.28]	+2.18	-0.66	+6.42
ViT-B/32	CUB-200	+2.16 [+2.06, +2.28]	+3.37	-1.34	+7.97
ViT-B/32	CIFAR-100	+2.12 [+2.03, +2.21]	+3.72	-1.23	+11.14
ViT-B/32	miniImageNet	+1.34 [+1.27, +1.41]	+2.32	-0.87	+7.11
RN50	CUB-200	+2.22 [+2.11, +2.34]	+3.74	-1.86	+9.21
RN50	CIFAR-100	+1.75 [+1.66, +1.84]	+2.89	-1.64	+9.70
RN50	miniImageNet	+1.75 [+1.67, +1.82]	+2.79	-1.00	+8.47

Table 34. Target-free QPR selector stability over five deterministic construction/selection/audit re-splits. Regret is the audit class-balanced gap to the best audit-safe grid point; audit results do not select a target tuple.

Dataset	Selected tuple frequencies $(\alpha, \tau_q, m_{\min}, \kappa)$	Unique	Audit pass	Mean regret	Max regret
CUB-200	(0.5,0.3,5,0.3)×1, (0.5,0.3,5,1)×1, (0.5,0.6,5,1)×1, (0.5,0.9,5,1)×1, (0.75,0.3,5,1)×1	5	5/5	0.02	0.04
CIFAR-100	(0.5,0.3,5,1)×3, (0.75,0.3,5,1)×2	2	5/5	0.02	0.05
miniImageNet	(0.5,0.3,5,1)×3, (0.5,0.3,5,0.3)×1, (0.5,0.6,5,1)×1	3	5/5	0.00	0.01

retains the PT-MAP transport/prototype-update core and its uniform marginal assumption; the released power transform is undefined for signed CLIP embeddings and is not applied.

The parent-normalized contrasts separate the stronger ShrinkSel starting head from QPR's contribution. In the original native-pipeline comparison reported here, QPR has the largest increment on CUB-200 and CIFAR-100, whereas TIM-GD has the largest increment on miniImageNet; QPR on ShrinkSel has the best absolute pipeline accuracy. The expanded two-head comparison in the camera-ready main paper also evaluates every comparator with ShrinkSel, where TIM-GD leads on CIFAR-100. The complete CUB trajectories retain both support-only references so the starting-head difference is visible.

The source transductive methods are commonly studied in balanced low-way episodes, whereas all-seen FSCIL com-

Table 35. Complete CUB-200 session trajectories for the parent-normalized query-assisted comparison. Cosine-NCM and ShrinkSel are support-only references; all other rows consume the complete unlabeled all-seen cohort. Sessions 1–10 average twenty paired support draws; session 0 is deterministic.

Method	Accuracy in each session (%) ↑											Avg. ↑
	0	1	2	3	4	5	6	7	8	9	10	
Cosine-NCM	81.63	79.14	77.49	74.25	73.26	71.69	70.81	70.04	68.53	68.72	68.48	73.09
ShrinkSel	84.25	81.71	80.23	77.36	76.50	75.08	74.38	73.87	72.62	72.83	72.72	76.50
ShrinkSel + QPR	84.25	82.48	81.38	78.64	78.06	76.74	76.43	76.08	74.94	75.20	75.25	78.13
ShrinkSel + QPR (self-excl.)	84.25	82.49	81.47	78.76	78.24	76.95	76.70	76.43	75.29	75.57	75.66	78.35
TIM-GD	80.55	78.92	77.14	73.98	73.03	71.44	70.85	70.27	68.82	69.00	68.83	72.99
LapShot-full	81.04	75.80	73.62	70.41	69.25	67.56	66.95	66.41	65.05	65.40	65.32	69.71
LapShot-core	81.53	79.02	77.38	74.15	73.20	71.63	70.78	70.01	68.51	68.69	68.45	73.03
ProtoLP	75.98	73.39	71.61	67.96	66.90	65.12	64.47	63.67	62.13	62.44	62.16	66.90
MAP-OT$_u$	67.74	65.32	63.28	60.71	59.34	57.20	55.90	54.31	52.31	52.13	51.24	58.13

pares 60–200 classes and retains reliable full-data base prototypes. The independent native-format check below verifies that the comparator implementations improve over cosine in the intended five-way regime. QPR's base–novel gate is undefined when every episode class is novel, so QPR is omitted from that check. A degenerate variant would have to delete that gate explicitly; we do not call it the proposed method.

Table 36. Native low-way sanity check on frozen CLIP ViT-B/16: balanced 5-way, 5-shot episodes with 15 queries per class. Values are mean accuracy ± 95% normal-approximation half-width over 1,000 deterministic episodes; this is an implementation check, not a reproduction of published accuracies. QPR is omitted because every class is novel, so its base–novel gate is undefined.

Method	CUB-200	CIFAR-100	miniImageNet
Cosine-NCM	94.94±0.32	88.13±0.45	97.70±0.13
TIM-GD	96.35±0.28	91.50±0.40	98.88±0.09
LaplacianShot-full	94.80±0.35	90.64±0.43	98.31±0.12
LaplacianShot-core	95.09±0.32	87.79±0.45	97.81±0.13
ProtoLP	94.91±0.32	89.20±0.44	97.95±0.13
MAP-OT$_u$ (PT-MAP core)	96.56±0.28	88.85±0.46	98.32±0.12

G.7. DTD transfer evaluation

We evaluate DTD with the same class-disjoint base selection and safety structure and twenty matched support draws. Results report the paired session-average difference and the final Full/Base/Novel changes separately.

QPR improves its matched parent by $+1.80$ session-average points with a 95% interval $[+1.57, +2.02]$. Its final novel gain is $+8.42$ points, accompanied by a -1.42-point base change. The parent ShrinkSel head is .23 point below shared LDA $[-.32, -.14]$. Thus the query-assisted update helps in this setting while the support-only transfer is adverse.

Table 37. DTD transfer evaluation on CLIP ViT-B/16. Paired session-average intervals use 20 supports; Δ_Q is QPR minus ShrinkSel.

Quantity	Value
ShrinkSel prototype / λ	resultant_debiased / 0.25
ShrinkSel base audit vs. LDA	+0.20
Target ShrinkSel − LDA [95% CI]	-0.23 [-0.32, -0.14]
QPR tuple	(0.5,0.3,5,0.3)
QPR base audit (balanced / anchor)	+0.76 / -0.18
Δ_Q session average [95% CI]	+1.80 [+1.57, +2.02]
Δ_Q final Full / Base / Novel	+2.77 / -1.42 / +8.42

H. Base-Session-Only Representation Control

For each standard benchmark, a ResNet-18 is initialized from scratch and optimized only on base-session training images. CIFAR-100 uses 200 epochs; CUB and miniImageNet use 120. Training uses SGD with Nesterov momentum .9, learning rate .1, cosine decay, weight decay 5×10^{-4}, label smoothing .1, and seed 1. CIFAR uses its standard 32-pixel crop/flip pipeline and a 3-by-3 first convolution without max pooling; CUB and miniImageNet use random resized crops and flips. After training, the 512-dimensional penultimate representation is frozen and normalized. Checkpoints record optimized base IDs and excluded novel IDs.

The full-mean replacement term is positive on every dataset, but no dataset passes the majority gate. CUB's matched-gap denominator is only 0.061 point, and its two-way-bootstrap 95% interval contains zero, so the ordered ratio is ill-conditioned and is omitted. The specified rule requires at least two stable passing datasets; because none qualifies, no support-only target-head comparison is run.

Table 38. Class-disjoint base-only representation control. All values are percentage points except r; "–" marks a ratio suppressed by the denominator-stability rule.

Dataset	A_n^K	A_n^{full}	A_n^{LP}	Mean Δ	r [95% CI]
CUB-200	18.54	27.58	18.60	+9.04	–
CIFAR-100	23.51	31.17	46.55	+7.67	0.33 [0.27, 0.40]
miniImageNet	15.00	24.22	34.87	+9.22	0.46 [0.39, 0.54]

I. Additional limitations of interpretation

The three CLIP encoders vary architecture within one pretraining family. MAE-ft and supervised ResNet-50 add families but share ImageNet-1K supervision, leaving pretraining objective and corpus exposure confounded. Aircraft adds a fourth domain only for CLIP ViT-B/16. The base-only control changes both exposure and representation quality, so it cannot separate their effects. Target-oracle shrinkage curves use labels for retrospective analysis and never guide selection. Resource measurements cover cached heads on one GPU; end-to-end latency also includes image decoding and feature extraction.

The primary base selectors emulate one new-class increment rather than the cumulative tm-class competition in later FSCIL sessions. The cumulative stress test retains the public count schedule but clips requests to the finite class-disjoint pseudo fold, so it cannot reproduce the largest target class counts. QPR is primarily same-cohort batch transduction. Its exact leave-one-query-out result establishes that each query can be held out from its own update, but the other cohort queries remain available. These are protocol boundaries rather than guarantees supplied by base-only safety checks.

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